

Total No. of Questions : 8]

SEAT No. :

PA-1622

[Total No. of Pages : 2

[5926]-256

T.E. (Computer)(Honors)

CYBER SECURITY

Information and Cyber Security

(2015 Pattern) (Semester - I) (310401)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.
- 5) Use of scientific calculator is permitted.

Q1) a) What is cryptographic hash function? How is it useful in cryptography? List different cryptographic hash functions. Explain in detail any one cryptographic hash function. **[8]**

b) Find the key exchanged between Alok and Bobby considering following data $n = 11$, $g = 5$, $x = 2$, $y = 3$. Find the value of A,B & key K. **[9]**

OR

Q2) a) What are steps carried out in diffie hellman algorithm? List uses, advantages and disadvantages of diffie hellman algo. **[8]**

b) What do you mean by Asymmetric cryptography algorithm? Explain RSA algorithm in detail. **[9]**

Q3) a) Describe different categories of cybercrime with example. **[9]**

b) Explain the process of risk identification and risk assessment. **[9]**

OR

Q4) a) What are the difference between quantitative and qualitative risk analysis with providing examples. **[9]**

b) What is cyber stalking? How to identify and detect cyber stalking. **[9]**

Q5) a) What is SSL? How does SSL works? Why is SSL important. **[8]**

b) Describe IPsec protocol with its components and security services. **[9]**

OR

P.T.O.

- Q6)** a) What is the firewall? How does it works & explain different types of firewalls. [8]
- b) What is email security and why it is necessary? Explain any one algorithm used for email security. [9]

- Q7)** a) What is malware? Enlist different types of malware what precaution needs to protect from malware. [9]
- b) What is computer worm or virus? How does computer virus spread? How to protect against computer virus and norms. [9]

OR

- Q8)** a) Enlist different types of IDS. Describe any one type of IDS in detail. [9]
- b) Define phishing. Explain phishing with types and examples. [9]

